ModelIt!

Teacher's Guide

How Does a Cut Heal Itself?

How Your Body Orchestrates Cell Division to Repair Damaged Tissue

Quick Reference

Grade Level	Grade 08L2
NGSS Standard	MS-LS1-4, MS-LS1-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: MS-LS1-4, MS-LS1-2

Use argument based on empirical evidence and scientific reasoning to support an explanation for how characteristic animal behaviors and specialized plant structures affect the probability of successful reproduction of animals and plants respectively. Develop and use a model to describe the function of a cell as a whole and ways the parts of cells contribute to the function.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a model showing how wound severity and blood supply affect cell division rate, new tissue formation, and overall healing progress.
Disciplinary Core Idea	**LS1.B Growth and Development of Organisms** — Animals engage in characteristic behaviors and have specialized body structures that increase the probability of successful reproduction and survival, including wound repair through cell division.
Crosscutting Concept	**Cause and Effect** — Students identify causal relationships between wound severity, blood supply, cell division rate, and healing outcomes — understanding that multiple causes interact to produce observed effects.

CAST Testing Alignment

MULTIPLE CHOICE:

Patient A has a moderate cut on their well-supplied forearm. Patient B has the same cut on their lower leg, where circulation is reduced due to diabetes. After 10 days, which patient will show more healing progress, and what is the primary reason?

CONSTRUCTED RESPONSE:

Using evidence from your model, explain why a deep wound with good blood supply might heal faster than a shallow wound with poor blood supply. Include the roles of cell division rate and new tissue formation in your explanation.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The human body is a remarkable self-repairing system. Every cut, scrape, and bruise triggers a precisely orchestrated cascade of cellular events that rebuild damaged tissue. Understanding how wounds heal reveals fundamental principles of cell biology — particularly cell division (mitosis) and how the body regulates growth.

The Four Phases of Wound Healing

Healing occurs in four overlapping phases: (1) Hemostasis — blood clotting stops the bleeding within minutes. (2) Inflammation — immune cells flood the area to fight infection and clear debris (hours to days). (3) Proliferation — cells divide rapidly to rebuild tissue, forming new skin, blood vessels, and connective tissue (days to weeks). (4) Remodeling — new tissue is reorganized and strengthened (weeks to months or years).

Cell Division: The Engine of Repair

Mitosis is the process by which one cell divides into two identical daughter cells. At a wound site, cells at the edge receive chemical signals (growth factors) that trigger them to divide much faster than normal. These new cells migrate into the wound, gradually filling the gap. The rate of cell division depends heavily on blood supply — dividing cells consume enormous amounts of oxygen and nutrients.

Blood Supply: The Lifeline

Blood delivers everything healing cells need: oxygen for cellular respiration (energy production), glucose and amino acids for building new cellular structures, growth factors that signal cells to divide, and white blood cells that fight infection. Areas with rich blood supply (face, tongue, scalp) heal remarkably fast. Areas with poor circulation (lower legs, feet) heal slowly — this is why diabetic foot wounds are a serious medical concern.

Why Some Wounds Scar

Small wounds can regenerate tissue that is virtually identical to the original. Large or deep wounds cannot wait for perfect regeneration — they fill the gap quickly with collagen-rich scar tissue. Scars are functional but lack the full complexity of original tissue (no hair follicles, sweat glands, or normal pigmentation). The body prioritizes speed of closure over perfection of repair.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify wound severity, blood supply, cell division rate, new tissue formation, and healing progress as system components.

E **Establish Relationships**
Students determine that wound severity sets the repair scale, blood supply fuels cell division, and cell division produces new tissue that drives healing progress.

V **Visualize & Model**
Students build the wound healing model showing how inputs (severity and blood supply) affect the cascade of cell division, tissue formation, and healing.

E **Evaluate Outcomes**
Students compare scenarios with different severity and blood supply levels to verify that blood supply is the rate-limiting factor for healing speed.

R **Revise & Extend**
Students add immune response, scar tissue formation, or growth factor signaling to model more complex healing dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"Every time you get a cut, billions of cells launch a repair mission. How does your body rebuild itself?"*

Watch For

Show microscopic wound healing image. Ask: What do you think happens in the first hour after a cut?

Slide 2: Learning Objectives

3 min

Say: *"Today we model how your body heals wounds through cell division."*

Watch For

Pre-teach wound severity (how bad), blood supply (delivery system), cell division (copying cells).

Slide 3: Big Question

3 min

Say: *"A paper cut heals in days. A deep gash takes weeks. Same process, different timelines. Why?"*

Watch For

Ask students to share experiences with cuts healing. What did they notice?

Slide 4: LEVER Framework

2 min

Say: *"We'll model the biological assembly line that rebuilds your tissue after injury."*

Watch For

Preview that healing is a cascade: severity sets the job, blood supply fuels the workers.

Slide 5: Activity 1: Components

8 min

Say: *"Wound severity and blood supply are our inputs. Cell division, tissue formation, and healing progress are outputs."*

Watch For

Sort components. Discuss why severity and blood supply are external (set by the injury and body).

Slide 6: Activity 2: Connections

8 min

Say: *"Blood supply fuels cell division. Cell division produces new tissue. New tissue drives healing progress."*

Watch For

Map the cascade. Highlight that blood supply is the rate-limiting factor.

Slide 7: Activity 3: Simulation

10 min

Say: *"Compare a minor cut with good blood supply vs. a deep wound with poor circulation. What happens?"*

Watch For

Students predict healing curves. Run scenarios. Identify the critical role of blood supply.

Slide 8: Discoveries

5 min

Say: *"Your body is an incredible self-repair system — but it needs fuel, time, and the right conditions."*

Watch For

Summarize key findings. Connect to real medical practice (wound care, diabetes management).

Slide 9: STEM Challenge

12 min

Say: *"You're the school nurse's science advisor. Design a treatment protocol backed by cell biology evidence."*

Watch For

Teams create evidence-based protocols. Present and defend with model data.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Wound Severity, Blood Supply
- **INTERNAL:** Cell Division Rate, New Tissue Formation, Healing Progress

Relationships

- **Wound Severity to Cell Division Rate:** POSITIVE (+) — More severe wounds trigger stronger chemical signals (growth factors) that increase the rate of cell division at the wound edge.
- **Wound Severity to Healing Progress:** NEGATIVE (-) — Greater wound severity means more tissue to replace, requiring more time to achieve complete healing — deeper wounds heal more slowly.
- **Blood Supply to Cell Division Rate:** POSITIVE (+) — Greater blood flow delivers more oxygen and nutrients to dividing cells, enabling faster mitosis and cell production.
- **Cell Division Rate to New Tissue Formation:** POSITIVE (+) — Faster cell division produces more new cells per day, accelerating the formation of replacement tissue.
- **New Tissue Formation to Healing Progress:** POSITIVE (+) — More new tissue filling the wound means greater healing progress — the wound closes and functional tissue is restored.

Key Discoveries

- Wound severity determines the scale of repair needed — more damaged tissue requires more cell divisions to replace
- Blood supply is critical because dividing cells need oxygen and nutrients delivered by blood to fuel mitosis
- Cell division rate increases near the wound edge as the body signals cells to multiply faster
- New tissue formation follows cell division — first new cells are produced, then they organize into functional tissue
- Healing progress depends on both factors: severe wounds with good blood supply heal faster than moderate wounds with poor circulation

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Scabs are the healing"	Scabs are blood clots that protect the wound, not the actual repair process. Real healing happens UNDERNEATH the scab as cells divide and form new tissue. In fact, keeping wounds moist (without a dry scab) often leads to faster healing because cells can migrate and divide more easily without having to dissolve dried material.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Wounds heal from the outside in"	Wounds actually heal from the edges inward and from the bottom up. Cells at the wound margin divide and migrate across the wound surface, while new blood vessels and connective tissue build up from below. This is why deep wounds take longer — there's more vertical depth to fill.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Your body just 'knows' when to heal"	Healing is triggered by specific chemical signals, not some mysterious awareness. When cells are damaged, they release growth factors and cytokines that signal neighboring cells to begin dividing. White blood cells arrive and release additional signals. This chemical cascade is automatic and measurable — it's biochemistry, not magic.